

CPTG and the Cosmic Standard Ruler

A Direct CMB-to-DESI Test of the Acoustic Projection

Carter L Glass Jr

Abstract

Baryon acoustic oscillations (BAO) are widely used as a cosmic standard ruler, but the observations do not measure an absolute ruler length directly. They measure cosmological distances relative to an acoustic scale whose dimensional calibration, in the standard Λ CDM interpretation, is supplied by an early-universe model. The inferred ruler is therefore interlocked with H_0 and with the same cosmological assumptions the ruler is later used to test. DESI DR2 has sharpened this issue: the BAO distance relation itself remains well described by flat Λ CDM, while combined CMB, BAO, and supernova analyses continue to show cross-dataset parameter stress and a dataset-dependent preference for evolving dark energy.

Curvature Polarization Transport Gravity (CPTG) uses a different construction. It begins from one fixed native geometric branch and its fixed acoustic projection. The analysis does not fit an independent dimensional DESI ruler. Instead, the measured CMB acoustic angle and the adopted Planck decoupling endpoint define the comparison boundary from which the fixed CPTG projection determines the dimensionless BAO normalization. The resulting coordinates are then tested directly in the variables reported by DESI.

This paper uses the DESI DR2 lower-redshift BAO covariance together with the joint Ly α AP+BAO measurement at $z_{\text{eff}} = 2.33$, which provides the high-redshift radial and transverse constraint. The CMB-derived CPTG normalization agrees with the normalization conditionally preferred by this DESI vector to 0.779σ . With no DESI-fitted ruler or normalization parameter, the 13-observable covariance test gives $\chi^2 = 13.6025$ for 13 degrees of freedom and $p = 0.4024$. The largest marginal individual residual is 1.61σ , and the result remains stable under redshift-block removal and separate radial and transverse tests.

The unprojected native branch fails the same DESI test even after one global normalization is optimized, giving $p = 1.27 \times 10^{-18}$. Within the comparison-coordinate and covariance scope tested here, the fixed CPTG acoustic projection therefore carries the CMB boundary into the DESI standard-ruler coordinates without a DESI-specific dimensional ruler calibration.

1 Introduction: The Cosmic Standard Ruler

Baryon acoustic oscillations are among the most precise geometrical observables in modern cosmology. A preferred separation is imprinted in the large-scale distribution of matter and can be measured at multiple redshifts using galaxies, quasars, and the Lyman- α forest. DESI reports this information through dimensionless radial, transverse, and isotropic distance ratios and therefore provides a high-precision test of the cosmological distance–redshift relation [2, 7].

The phrase *standard ruler*, however, can hide an important distinction. BAO observations do not directly return an absolute length in megaparsecs. They measure distances relative to an acoustic scale. In the standard Λ CDM interpretation, the dimensional ruler is the drag-epoch sound horizon r_d , whose calibration depends on the assumed early-universe expansion and plasma composition. The ruler used to infer late-time distances is therefore not independent of the cosmological model used to calibrate it [1, 2].

This creates a structural interlock between the acoustic ruler and the expansion scale. BAO constrain combinations such as D_M/r_d , D_H/r_d , and, in inverse-distance-ladder applications, $H_0 r_d$. Raising a late-time value of H_0 while preserving the observed BAO ratios generally requires a correspondingly smaller calibrated r_d . Recent inverse-distance-ladder analyses using DESI BAO and supernova distances continue to recover H_0 values near the CMB-calibrated scale rather than the local distance-ladder scale [8]. The standard ruler is therefore part of the cross-epoch consistency problem rather than an external absolute yardstick.

DESI has also sharpened the distinction between observable-space agreement and parameter-space closure. The DR2 BAO distance relation is well described by flat Λ CDM, but the BAO-preferred parameters are in mild tension with CMB-inferred parameters [2]. Allowing an evolving equation of state improves combined fits, with the significance depending on the dataset combination [2, 7]. These results do not establish evolving dark energy; they show that the interpretation of the ruler within one common standard-model parameter set remains an active cross-dataset question.

An independent statistical analysis of the DESI DR2 vector illustrates the same distinction from another direction, reporting a smaller discrepancy in the measured BAO data space than in the derived $(\Omega_m, H_0 r_d)$ parameter space [9]. This result is not a DESI Collaboration claim and is not used in the likelihood developed below; it is cited only to motivate the distinction between matching the observed distance vector and closing a model’s inferred parameter set.

Three-dimensional BAO measurements also require a fiducial cosmology for data compression from angles and redshifts into comoving coordinates. This does not make the standard DESI analysis circular: Alcock–Paczynski scaling is designed to recover departures from the fiducial geometry, and DESI validation studies find fiducial-cosmology systematics at approximately the 0.1% level for the tested BAO dilation parameters [10]. The issue for a nonstandard framework is therefore not to reject the DESI coordinates, but to state explicitly how its own geometry is translated into the coordinates in which the survey likelihood is reported.

Curvature Polarization Transport Gravity (CPTG) makes that translation without fitting an independent dimensional DESI ruler. CPTG begins from one native geometric branch and a fixed acoustic projection of that branch [3, 4]. Once the relevant beginning and ending comparison coordinates are specified, the projected geometry determines the corresponding observational coordinate. The CMB acoustic angle and the adopted decoupling endpoint provide the upstream comparison boundary used in this paper; the CPTG acoustic expansion scale is fixed independently by the native branch.

Methodologically, this paper is a direct extension of the CPTG Hubble-Tension Bridge construction [5, 4]. Both analyses begin from the same native geometric history, carry that history through the fixed acoustic projection, and compare observational endpoints as coordinate readouts of one underlying CPTG evolution rather than as independently fitted cosmologies. The distinction here is that the standard-ruler test pushes that same projection methodology beyond endpoint comparison and into the redshift-dependent DESI BAO distance coordinates, where radial and transverse responses can be tested simultaneously with their covariance.

For anisotropic BAO measurements, the principal reported coordinates are

$$\frac{D_M(z)}{r}, \quad \frac{D_H(z)}{r}. \quad (1)$$

Here D_M is the transverse distance coordinate and

$$D_H(z) = \frac{c}{H(z)} \quad (2)$$

is the radial Hubble-distance coordinate. Isotropic measurements are commonly expressed through

$$\frac{D_V(z)}{r}, \quad D_V(z) = \left[z D_M(z)^2 D_H(z) \right]^{1/3}. \quad (3)$$

The symbol r denotes the ruler coordinate used by the survey. In conventional DESI notation this is r_d . CPTG does not require the physical interpretation of that denominator to be identical to the standard-model drag-epoch sound horizon; the test is whether CPTG supplies the same reported dimensionless coordinates.

The primary question of this paper is therefore whether one fixed CPTG native branch, through its already-defined acoustic projection, can reproduce the observed CMB-to-DESI standard-ruler coordinates without a DESI-specific ruler or normalization fit.

The controlling DESI vector combines the public DR2 lower-redshift BAO measurements with the joint Ly α AP+BAO result at $z_{\text{eff}} = 2.33$ [2, 7]. The CPTG acoustic exponent and CMB-side normalization are fixed upstream. No DESI-specific value of H_0 , r , or $A = (c/H_0)/r$ is fitted in the controlling test. Robustness is then assessed by deleting complete redshift blocks, separating radial and transverse sectors, partitioning the redshift range, and comparing the projected acoustic branch with the unprojected native π -branch as a negative control.

2 CPTG in Cosmological Context

Curvature Polarization Transport Gravity is a geometric framework for gravitational, cosmological, structural, and nuclear-response phenomena [3]. The present paper uses only its cosmological branch architecture.

The first distinction needed here is between a *native branch* and an *observational projection*. The native branch is the underlying CPTG cosmological geometry. An observational projection is a derived representation of that same native history in the coordinate language appropriate to a particular measurement channel. The acoustic branch used for CMB and BAO comparisons is one such projection. It is not an independently fitted expansion history and does not introduce a second native cosmology.

The construction may be summarized schematically as

$$\mathfrak{B}_\pi \xrightarrow{\mathcal{P}_{ac}} \mathfrak{B}_{ac} \xrightarrow{\text{observational coordinate map}} \{\text{CMB, BAO}\}, \quad (4)$$

where $\mathfrak{B}_\pi$ denotes the native geometric branch and $\mathfrak{B}_{ac}$ its acoustic projection. Equation (4) is a dependency chain, not material transport between separate universes.

The native and projected descriptions are compared at a common geometric phase. Writing

$$N_\pi \equiv \ln a, \quad (5)$$

the same value of N_π identifies the same geometric stage of the underlying history. A projected branch may assign a different accumulated comparison-time coordinate to that stage. For any branch b whose rate history is defined,

$$H_b(N_\pi) dt_b = dN_\pi. \quad (6)$$

Thus equal geometric phase does not require equal accumulated branch-clock coordinates. In this paper the common-phase relation is used only to keep the native and acoustic descriptions synchronized geometrically; it is not used to introduce an additional time law for the BAO ruler.

The second distinction is between theory quantities and observational comparison coordinates. CPTG fixes the native branch and its acoustic projector upstream. CMB and BAO quantities then enter according to their observational roles. In particular, the observed CMB angular acoustic scale $\theta_\star$ is used as an observational comparison scaffold. It is not reclassified as a native CPTG invariant or presented here as a theory-fixed quantity derived independently of the declared observational scaffold. Likewise, DESI quantities such as D_M/r , D_H/r , and D_V/r are treated as the reported coordinates in which the fixed CPTG acoustic projection is tested.

This hierarchy is important for interpreting the result. Agreement with DESI does not mean that DESI defines the CPTG acoustic branch. The inference direction is the

opposite:

$$\begin{aligned}
&\text{native CPTG geometry} \downarrow \\
&\text{acoustic projection} \downarrow \\
&\text{CMB comparison coordinate} \downarrow \\
&\longrightarrow \text{DESI BAO coordinates}
\end{aligned} \tag{7}$$

The DESI likelihood is therefore downstream of the theory construction. The controlling test in this paper keeps the upstream CPTG quantities fixed and asks whether the resulting BAO vector is accepted by the DESI covariance.

One input distinction is important for everything that follows. The Planck base- Λ CDM inference $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is not a CPTG input. CPTG instead carries forward its independently derived acoustic value,

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

which is fixed by the native branch and acoustic projection before any Planck comparison is made. The Planck value is used only as an external observational comparison. The standard-ruler calculation itself uses the CPTG-derived acoustic value together with the measured acoustic angle and adopted decoupling redshift identified in Section 6. CPTG is treated here as a fixed geometric mapping: once the relevant beginning coordinate, ending coordinate, and projection are specified, the corresponding comparison quantity follows without introducing a tunable CDM calibration sector.

The remainder of the derivation now proceeds from the native branch itself.

3 The Native Geometric π -Branch

The standard-ruler calculation begins from the CPTG native cosmological branch rather than from an observationally fitted BAO background. This ordering is important. The native branch fixes the geometric expansion history that is later translated into the acoustic comparison coordinates used by the CMB and DESI; the observational data do not define the native branch itself [3, 4].

The native branch exponent is

$$p_C = \pi, \tag{8}$$

with dimensionless weights

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}. \tag{9}$$

The corresponding dimensionless native expansion function is

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \tag{10}$$

or, equivalently,

$$E_\pi(z)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi(1+z)^\pi. \quad (11)$$

The normalization satisfies

$$E_\pi(1) = 1, \quad (12)$$

so the native rate is

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a). \quad (13)$$

The present native rate normalization is not chosen by interpolating between a CMB-inferred and local distance-ladder value. The acceleration anchor used here is an established upstream CPTG quantity from the parent geometric construction [3, 4]; it is recorded rather than re-derived in this standard-ruler paper:

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} \text{ m s}^{-2}, \quad (14)$$

which gives the asymptotic rate scale

$$H_\infty = \frac{a_{\pi,0}}{c} = 1.85739429003 \times 10^{-18} \text{ s}^{-1} = 57.3131992073 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (15)$$

where the final equality is the standard conversion from s^{-1} to $\text{km s}^{-1} \text{ Mpc}^{-1}$.

Using the native late/geometric weight $\mathcal{B}_\pi$, the present native normalization is therefore

$$H_0^{(\pi)} = \frac{H_\infty}{\sqrt{\mathcal{B}_\pi}} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (16)$$

The dependency chain is thus

$$a_{\pi,0} \longrightarrow H_\infty \longrightarrow H_0^{(\pi)} \longrightarrow H_\pi(a), \quad (17)$$

with no CMB or BAO normalization inserted into Eqs. (14)–(17).

For the present paper, Eqs. (10)–(13) are the complete native cosmological input required before the acoustic projection is applied. They should not be read as an independent fit to the DESI distance relation. The role of the native branch is upstream: it supplies the single geometric state from which the acoustic comparison branch is constructed.

A useful way to expose the part of the native state that is changed by the acoustic projection is to separate the constant and nonconstant pieces of Eq. (10). Define

$$X_\pi(a) \equiv E_\pi(a)^2 - \mathcal{B}_\pi = \mathcal{T}_\pi a^{-\pi}. \quad (18)$$

The acoustic projection acts on this nonconstant native contribution. It changes the response exponent while preserving the inherited native coefficient, as shown next.

4 The Exact Acoustic Projection

The acoustic comparison branch is derived from the native state rather than initialized with an independent background amplitude. Its exponent is an established upstream CPTG projection quantity [4]; it is not fitted to Planck or DESI in this paper:

$$p_{\text{ac}} = 3 - \frac{\pi}{100} = 2.968584073464102, \quad (19)$$

and the corresponding projection ratio is

$$\mathcal{G}_T \equiv \frac{p_{\text{ac}}}{\pi} = 0.9449296585513721. \quad (20)$$

Its square-root response factor is

$$\sqrt{\mathcal{G}_T} = 0.9720749243506758. \quad (21)$$

The exact normalized acoustic projector is

$$\mathcal{P}_{\text{ac}}(X) \equiv \mathcal{T}_{\pi}^{1-\mathcal{G}_T} X^{\mathcal{G}_T}. \quad (22)$$

Applying Eq. (22) to the native contribution in Eq. (18) gives

$$X_{\text{ac}}(a) = \mathcal{P}_{\text{ac}}[X_{\pi}(a)] \quad (23)$$

$$= \mathcal{T}_{\pi}^{1-\mathcal{G}_T} \left(\mathcal{T}_{\pi} a^{-\pi} \right)^{\mathcal{G}_T} \quad (24)$$

$$= \mathcal{T}_{\pi} a^{-\pi \mathcal{G}_T} \quad (25)$$

$$= \mathcal{T}_{\pi} a^{-p_{\text{ac}}}. \quad (26)$$

The projected acoustic expansion function is therefore

$$E_{\text{ac}}(a)^2 = \mathcal{B}_{\pi} + \mathcal{T}_{\pi} a^{-p_{\text{ac}}}, \quad (27)$$

or

$$E_{\text{ac}}(z)^2 = \mathcal{B}_{\pi} + \mathcal{T}_{\pi} (1+z)^{p_{\text{ac}}}. \quad (28)$$

Equation (26) contains the essential normalization result. The acoustic projection changes the exponent from π to p_{ac} but preserves the native coefficient $\mathcal{T}_{\pi}$ exactly. The acoustic background therefore does not require a second primordial normalization to be fitted to CMB or BAO data. Within the declared local monomial class $\mathcal{P}(X) = AX^{\gamma}$, the simultaneous requirements of exponent transfer $\pi \mapsto p_{\text{ac}}$ and exact inheritance of the coefficient $\mathcal{T}_{\pi}$ fix

$$\gamma = \mathcal{G}_T, \quad A = \mathcal{T}_{\pi}^{1-\mathcal{G}_T}, \quad (29)$$

which recovers Eq. (22). This is a uniqueness statement within that registered monomial class; no broader uniqueness over arbitrary nonlinear projector functionals is required here.

The acoustic present-rate comparison coordinate follows from the fixed rate projection

$$H_{0,\text{ac}}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (30)$$

The complete acoustic rate history used in this paper is then

$$H_{\text{ac}}(a) = H_{0,\text{ac}}^{\text{CPTG}} E_{\text{ac}}(a). \quad (31)$$

Equations (19)–(31) are fixed before the DESI comparison is performed.

4.1 CMB-facing rate comparison

Before constructing the standard-ruler coordinate, it is useful to verify that the acoustic endpoint lies in the observed CMB-facing range. Planck 2018 reports the base- Λ CDM CMB-inferred value [1]

$$H_0^{\text{Planck}} = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (32)$$

The residual is

$$\Delta H_0 \equiv H_{0,\text{ac}}^{\text{CPTG}} - H_0^{\text{Planck}} = +0.0777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (33)$$

so, using the published Planck base- Λ CDM inference uncertainty,

$$\frac{\Delta H_0}{\sigma_{H_0}^{\text{Planck}}} = \frac{0.0777967351}{0.5} = 0.1556\sigma. \quad (34)$$

Thus the theory-fixed acoustic endpoint differs from the Planck CMB-inferred endpoint by only 0.156σ . The comparison is not itself the standard-ruler closure developed below, but it establishes that the acoustic rate normalization generated by the same projector is observationally consistent before any DESI information is introduced.

The next step is to translate this fixed acoustic rate history into the radial, transverse, and volume-averaged coordinates in which the cosmic standard ruler is actually measured.

5 From the Acoustic Projection to the Cosmic Standard Ruler

The acoustic projection fixes a redshift-dependent rate history. The cosmic standard ruler is not observed as that rate directly; it is observed through distance coordinates constructed from the rate. This section makes that translation explicit.

The distance relations below are used as the DESI/BAO *comparison-coordinate adapter*. They specify how the CPTG projected rate history is expressed in the compressed radial and transverse variables reported by the survey, following the CPTG comparison-coordinate construction [4]. They are not introduced as independent native CPTG

governing equations, as an FLRW background law for CPTG, or as a replacement background cosmology.

For the projected acoustic branch,

$$H_{\text{ac}}(z) = H_{0,\text{ac}}^{\text{CPTG}} E_{\text{ac}}(z). \quad (35)$$

The corresponding radial Hubble-distance coordinate is

$$D_{H,\text{ac}}(z) = \frac{c}{H_{\text{ac}}(z)} = \frac{c}{H_{0,\text{ac}}^{\text{CPTG}} E_{\text{ac}}(z)}. \quad (36)$$

The transverse comoving-distance coordinate is

$$D_{M,\text{ac}}(z) = \frac{c}{H_{0,\text{ac}}^{\text{CPTG}}} \int_0^z \frac{dz'}{E_{\text{ac}}(z')}. \quad (37)$$

For convenience define

$$I_{\text{ac}}(z) \equiv \int_0^z \frac{dz'}{E_{\text{ac}}(z')}. \quad (38)$$

Then

$$D_{M,\text{ac}}(z) = \frac{c}{H_{0,\text{ac}}^{\text{CPTG}}} I_{\text{ac}}(z). \quad (39)$$

The isotropic volume-averaged coordinate is

$$D_{V,\text{ac}}(z) = \left[z D_{M,\text{ac}}(z)^2 D_{H,\text{ac}}(z) \right]^{1/3}. \quad (40)$$

These definitions already explain why the standard ruler need not appear as one rigid coordinate under every projection. The radial distance is local in the projected rate at the observed redshift, whereas the transverse distance integrates the projected rate over the full path from the observer to that redshift. The two coordinates therefore respond differently whenever the native-to-acoustic rate ratio varies with phase.

To make the relation precise, define the local acoustic response ratio

$$\Lambda_{\text{ac}}(z) \equiv \frac{H_{\text{ac}}(z)}{H_{\pi}(z)} = \sqrt{\mathcal{G}_T} \frac{E_{\text{ac}}(z)}{E_{\pi}(z)}. \quad (41)$$

The radial distance then obeys the exact coordinate identity

$$\frac{D_{H,\text{ac}}(z)}{D_{H,\pi}(z)} = \Lambda_{\text{ac}}(z)^{-1}. \quad (42)$$

The transverse relation is instead

$$\frac{D_{M,\text{ac}}(z)}{D_{M,\pi}(z)} = \frac{1}{\sqrt{\mathcal{G}_T}} \frac{\int_0^z dz' / E_{\text{ac}}(z')}{\int_0^z dz' / E_{\pi}(z')}. \quad (43)$$

Equations (42) and (43) do not introduce another CPTG dynamical law. They are derived coordinate consequences of the existing acoustic projection together with the registered DESI comparison-coordinate mapping used in this paper [4]. In particular, the apparent “bending” of a ruler representation is not interpreted here as physical evolution of an independent rod. It is the difference between a local projected rate coordinate and an integrated projected distance coordinate.

The same point can be stated directly in the observable BAO ratios. Suppose a survey reports a dimensionless coordinate

$$q_X(z) = \frac{D_X(z)}{r}, \quad (44)$$

for $X \in \{H, M, V\}$. A representation of the same dimensionless observation in another comparison coordinate may change both the distance and the dimensional ruler assigned to that coordinate while leaving q_X unchanged. The invariant observational object is the reported ratio, not an independently postulated CPTG ruler-evolution law.

The controlling calculation below therefore does not attempt to evolve a dimensional ruler through redshift. Instead, the fixed acoustic branch is expressed directly in the same dimensionless coordinates reported by DESI. The remaining task is to determine the one upstream acoustic normalization required to perform that comparison. The CMB angular acoustic scale supplies that observational anchor.

6 The CMB Acoustic Anchor and Its Uncertainty

The CMB standard-ruler construction enters this paper through two Planck 2018 comparison quantities: the directly constrained angular acoustic scale and the Planck-derived photon-decoupling endpoint. For the Planck TT,TE,EE+lowE+lensing combination used here,

$$100\theta_{\star}^{\text{obs}} = 1.04110 \pm 0.00031, \quad (45)$$

and

$$z_{\star}^{\text{Pl}} = 1089.92 \pm 0.25. \quad (46)$$

The quoted intervals are the Planck 68% comparison uncertainties [1]. The acoustic angle is directly constrained by the CMB likelihood, whereas $z_{\star}^{\text{Pl}}$ is a derived decoupling coordinate. Neither quantity is reclassified as a native CPTG invariant.

For the fixed acoustic shape in Eq. (28), the dimensionless distance integral to the adopted last-scattering phase is

$$I_{\star} \equiv I_{\text{ac}}(z_{\star}) = \int_0^{z_{\star}} \frac{dz}{E_{\text{ac}}(z)} = 3.213146044210494. \quad (47)$$

The fully acoustic comoving-distance coordinate is therefore

$$\chi_{\star, \text{ac}} = \frac{c}{H_{0, \text{ac}}^{\text{CPTG}}} I_{\star}. \quad (48)$$

A dimensional-coordinate convention must be stated explicitly here. The parent CPTG comparison-coordinate reference uses the same acoustic integrand with the native dimensional prefactor,

$$\chi_{\star}^{[\pi]} = \frac{c}{H_0^{(\pi)}} I_{\star}, \quad r_{\text{ac}}^{[\pi]} = \theta_{\star}^{\text{obs}} \chi_{\star}^{[\pi]} = 144.4716 \text{ Mpc}, \quad (49)$$

as defined in Ref. [4]. The present paper instead uses the fully acoustic dimensional representation of Eq. (48); throughout this paper $r_{\text{ac}} \equiv r_{\text{ac}}^{[\text{ac}]}$ denotes that representation. The two dimensional coordinates are not identified with one another. Their distinction does not alter the dimensionless CMB-to-DESI test, because in either convention the same dimensional prefactor appears in both the distance and ruler and cancels exactly:

$$\frac{c/H_0}{\theta_{\star}^{\text{obs}}(c/H_0)I_{\star}} = \frac{1}{\theta_{\star}^{\text{obs}}I_{\star}}. \quad (50)$$

The CMB angular acoustic coordinate then implies

$$r_{\text{ac}} = \theta_{\star}^{\text{obs}} \chi_{\star, \text{ac}} = \theta_{\star}^{\text{obs}} \frac{c}{H_{0, \text{ac}}^{\text{CPTG}}} I_{\star}. \quad (51)$$

Using the central values in Eqs. (45) and (46) gives

$$r_{\text{ac}} = 148.621869963 \text{ Mpc}. \quad (52)$$

For direct BAO comparison it is more useful to divide the distance scale by the ruler before carrying the result downstream. Define

$$A_{\text{CPTG}} \equiv \frac{c/H_{0, \text{ac}}^{\text{CPTG}}}{r_{\text{ac}}}. \quad (53)$$

Substituting Eq. (51) into Eq. (53) eliminates both the dimensional rate normalization and c :

$$A_{\text{CPTG}} = \frac{1}{\theta_{\star}^{\text{obs}} I_{\star}} = 29.8935221443. \quad (54)$$

This cancellation is important. Once the acoustic shape is fixed, the dimensionless BAO normalization is determined by the CMB angular coordinate and the dimensionless acoustic distance integral. No DESI value of H_0 or r is required.

6.1 Propagation of the CMB comparison uncertainty

The CPTG quantities p_{ac} , $\mathcal{G}_T$, and $H_{0, \text{ac}}^{\text{CPTG}}$ are treated as fixed theory quantities in the present test. The comparison uncertainty attached to Eq. (54) therefore comes from the Planck quantities $\theta_{\star}^{\text{obs}}$ and $z_{\star}^{\text{Pl}}$.

With

$$A = [\theta_\star I_{\text{ac}}(z_\star)]^{-1}, \quad (55)$$

the relevant derivatives are

$$\frac{\partial A}{\partial \theta_\star} = -\frac{A}{\theta_\star}, \quad (56)$$

$$\frac{\partial A}{\partial z_\star} = -\frac{A}{I_\star E_{\text{ac}}(z_\star)}. \quad (57)$$

For the comparison-level propagation used here, the quoted Planck uncertainties are treated as independent. This is an approximation because both quantities originate from the same CMB analysis, but the propagated uncertainty is overwhelmingly controlled by $\theta_\star$. The separate 1σ contributions to A_{CPTG} are approximately 0.0089012 from $\theta_\star$ and 0.0001277 from $z_\star^{\text{Pl}}$. The result is therefore insensitive to any plausible correlation at the precision relevant to this comparison. With that declared approximation,

$$\sigma_{A,\text{CMB}}^2 = \left(\frac{\partial A}{\partial \theta_\star} \sigma_{\theta_\star} \right)^2 + \left(\frac{\partial A}{\partial z_\star} \sigma_{z_\star} \right)^2. \quad (58)$$

Numerically,

$$A_{\text{CPTG}} = 29.8935221443 \pm 0.0089020704. \quad (59)$$

The corresponding dimensional acoustic comparison coordinate is

$$r_{\text{ac}} = 148.621869963 \pm 0.044258497 \text{ Mpc}, \quad (60)$$

where the uncertainty again reflects only the propagated Planck comparison inputs in Eqs. (45) and (46). It is not a theory-parameter error budget.

The CMB step is therefore complete before DESI is introduced: the theory fixes the acoustic shape, Planck supplies the declared observational angular scaffold, and together they determine the dimensionless normalization and its comparison-level uncertainty.

7 Direct CPTG Prediction in DESI BAO Coordinates

DESI reports the standard ruler through dimensionless distance ratios conventionally written with the drag-epoch denominator r_d . The CPTG construction must therefore distinguish a survey coordinate label from a claim about the native physical mechanism.

The CMB-side quantity r_{ac} derived in Section 6 is a CPTG acoustic *comparison coordinate*. It is constructed from the measured CMB acoustic angle at the adopted decoupling endpoint. It is not asserted to be the standard-model physical drag horizon, and this paper does not set $r_{\text{ac}} = r_d$ as a microscopic identity. Instead, DESI's r_d -labelled ratios define the coordinate system in which the survey likelihood is published. The test is whether the fixed CPTG acoustic projection supplies those same dimensionless coordinates.

Because Eq. (54) supplies the required dimensionless normalization, the CPTG prediction is

$$\left(\frac{D_M(z)}{r}\right)_{\text{CPTG}} = A_{\text{CPTG}} I_{\text{ac}}(z). \quad (61)$$

The radial coordinate is

$$\left(\frac{D_H(z)}{r}\right)_{\text{CPTG}} = \frac{A_{\text{CPTG}}}{E_{\text{ac}}(z)}. \quad (62)$$

The isotropic coordinate is

$$\left(\frac{D_V(z)}{r}\right)_{\text{CPTG}} = A_{\text{CPTG}} \left[\frac{z I_{\text{ac}}(z)^2}{E_{\text{ac}}(z)} \right]^{1/3}. \quad (63)$$

No DESI-specific dimensional ruler or normalization appears in Eqs. (61)–(63). Their dependency chain is

$$\mathfrak{B}_\pi \longrightarrow \mathcal{P}_{\text{ac}} \longrightarrow E_{\text{ac}}(z) \longrightarrow (\theta_\star^{\text{obs}}, z_\star^{\text{Pl}}) \longrightarrow A_{\text{CPTG}} \longrightarrow \left\{ \frac{D_M}{r}, \frac{D_H}{r}, \frac{D_V}{r} \right\}_{\text{DESI}}. \quad (64)$$

At $z_{\text{eff}} = 2.33$, the DESI joint Ly α AP+BAO analysis reports

$$\frac{D_H}{r_d} = 8.600 \pm 0.066, \quad \frac{D_M}{r_d} = 39.32 \pm 0.33, \quad (65)$$

with correlation coefficient

$$\rho(D_M/r_d, D_H/r_d) = 0.225 \quad (66)$$

[7]. These measurements define the high-redshift block of the controlling comparison.

Figure 1 shows the fixed CPTG curves against the DESI vector. The plotted error bars are marginal 1σ uncertainties. The decision statistic in Section 8 uses the full covariance within every redshift block, including the published Ly α correlation.

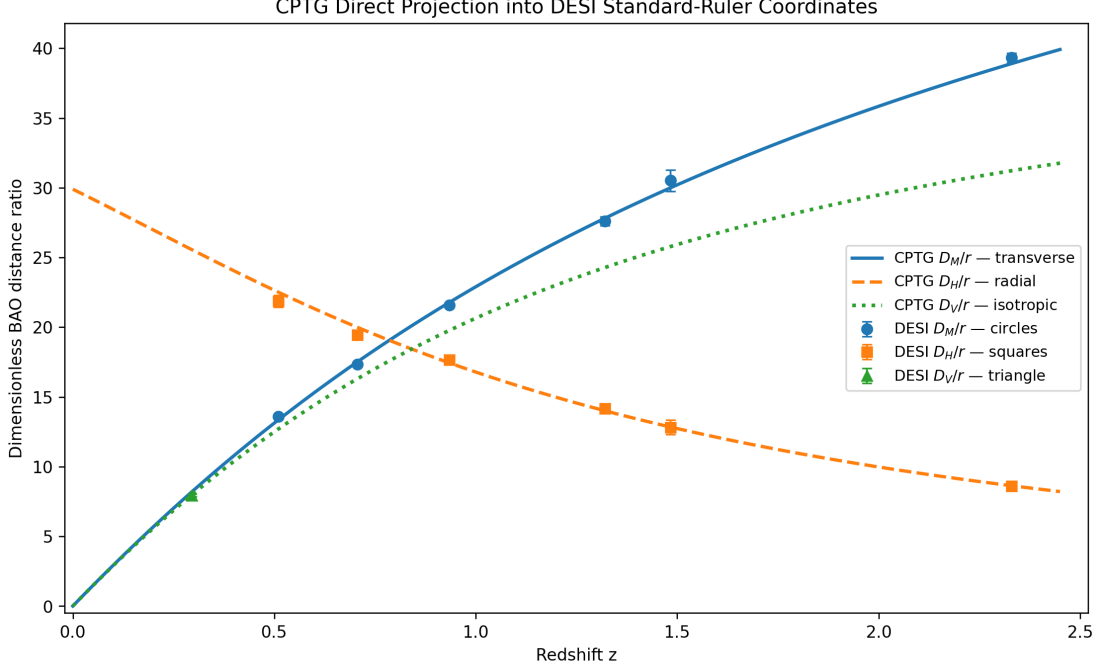


Figure 1: The fixed CPTG acoustic projection expressed directly in the DESI standard-ruler coordinates. The solid curve is the transverse prediction D_M/r , the dashed curve is the radial prediction D_H/r , and the dotted curve is the isotropic prediction D_V/r . DESI transverse measurements are shown with circles, radial measurements with squares, and the isotropic measurement with a triangle; each marker uses the same category color as its corresponding CPTG curve. The opposite trend of the radial curve is physical rather than graphical: $D_H/r = A_{\text{CPTG}}/E_{\text{ac}}(z)$ decreases as the projected expansion rate increases with redshift, whereas $D_M/r = A_{\text{CPTG}}I_{\text{ac}}(z)$ accumulates along the line of sight and therefore increases. Error bars are marginal 1σ uncertainties; the likelihood uses the corresponding covariance blocks.

The visual comparison is not the decision statistic. The controlling result is the covariance test developed next.

8 DESI Block-Covariance Test

The controlling data vector contains 13 observables. The first 11 are the public DESI DR2 lower-redshift BAO measurements through $z = 1.484$ [2]. The final two are the joint $\text{Ly}\alpha$ AP+BAO measurements at $z_{\text{eff}} = 2.33$ from Section 7 [7].

The covariance consists of the public DESI DR2 lower-redshift block together with the $\text{Ly}\alpha$ AP+BAO 2×2 high-redshift block,

$$\mathbf{C}_{\text{Ly}\alpha} = \begin{pmatrix} 0.066^2 & 0.225(0.066)(0.33) \\ 0.225(0.066)(0.33) & 0.33^2 \end{pmatrix}, \quad (67)$$

ordered as $(D_H/r_d, D_M/r_d)$. No additional cross-block covariance is introduced. Thus $\mathbf{C}$ is an analysis-level block covariance assembled from two published DESI likelihood components; it is not presented as a single DESI-released 13×13 covariance matrix.

Let the 13-element data vector be $\mathbf{d}$, with covariance $\mathbf{C}$. For the fixed acoustic exponent,

define the dimensionless shape vector

$$\mathbf{f}_{\text{ac}} = [f_i(p_{\text{ac}})]_{i=1}^{13}. \quad (68)$$

The zero-DESI-fit CPTG prediction is

$$\mathbf{m}_{\text{CPTG}} = A_{\text{CPTG}} \mathbf{f}_{\text{ac}}. \quad (69)$$

The residual vector and covariance statistic are

$$\Delta = \mathbf{m}_{\text{CPTG}} - \mathbf{d} \quad (70)$$

and

$$\chi_{\text{CPTG}}^2 = \Delta^T \mathbf{C}^{-1} \Delta. \quad (71)$$

No DESI quantity is adjusted in Eq. (71):

$$N_{\text{DESI fit}} = 0. \quad (72)$$

For the 13-observable vector,

$$\chi_{\text{CPTG}}^2 = 13.6024620904, \quad \nu = 13, \quad (73)$$

giving

$$\frac{\chi^2}{\nu} = 1.0463432377, \quad p = 0.4024238163. \quad (74)$$

The fixed CPTG vector remains statistically acceptable under the Ly α AP+BAO high-redshift constraint.

8.1 Comparison with the DESI-preferred normalization

As a diagnostic only, hold the CPTG acoustic shape fixed and allow a single common normalization,

$$\mathbf{m}(A) = A \mathbf{f}_{\text{ac}}. \quad (75)$$

The generalized-least-squares solution is

$$A_{\text{DESI}} = \frac{\mathbf{f}_{\text{ac}}^T \mathbf{C}^{-1} \mathbf{d}}{\mathbf{f}_{\text{ac}}^T \mathbf{C}^{-1} \mathbf{f}_{\text{ac}}} = 29.8295509892, \quad (76)$$

with conditional uncertainty

$$\sigma_{A,\text{DESI}} = \left(\mathbf{f}_{\text{ac}}^T \mathbf{C}^{-1} \mathbf{f}_{\text{ac}} \right)^{-1/2} = 0.0816008671. \quad (77)$$

Combining this with the CMB-side comparison uncertainty in Eq. (59),

$$\Delta A = A_{\text{CPTG}} - A_{\text{DESI}} = +0.0639711551 \quad (78)$$

and

$$\sigma_{\Delta A} = 0.0820850069. \quad (79)$$

Therefore

$$\frac{\Delta A}{\sigma_{\Delta A}} = 0.7793\sigma. \quad (80)$$

The DESI constraint leaves the CMB-derived CPTG normalization within 1σ .

For an intuitive dimensional representation only, the DESI-preferred equivalent ruler is

$$r_{\text{DESI|CPTG}} = 148.940597948 \pm 0.407437643 \text{ Mpc}, \quad (81)$$

compared with

$$r_{\text{ac}} = 148.621869963 \pm 0.044258497 \text{ Mpc}. \quad (82)$$

Their difference is

$$\Delta r = -0.318727985 \pm 0.409834414 \text{ Mpc} = -0.778\sigma. \quad (83)$$

This dimensional representation does not alter the coordinate-status distinction in Section 7: r_{ac} is the CPTG acoustic comparison coordinate, not a claimed microscopic identity with the standard-model drag horizon.

Allowing the one DESI normalization to float gives

$$\chi_{\text{free}}^2 = 12.9878815030 \quad (84)$$

for 12 degrees of freedom. Fixing the normalization upstream therefore costs only

$$\Delta\chi^2 = 0.6145805874, \quad (85)$$

consistent with the 0.779σ normalization displacement. The next section tests whether the closure remains stable when the DESI information is partitioned.

9 Robustness, Residuals, and Negative Controls

The zero-fit result would be less informative if it were carried by one redshift block or one distance sector. All robustness calculations therefore keep the CPTG prediction fixed and change only the subset of the DESI covariance being evaluated.

9.1 Leave-one-redshift-block-out stability

Each redshift block was removed in turn without changing p_{ac} , A_{CPTG} , or any other CPTG quantity. The resulting probabilities span

$$0.290 \leq p_{\text{LOO}} \leq 0.685. \quad (86)$$

The largest reduction in the full statistic occurs when the $z = 0.706$ block is omitted:

$$\Delta\chi_{\text{LOO}}^2(z = 0.706) = 5.2852. \quad (87)$$

The remaining vector then gives

$$\chi^2 = 8.3173, \quad \nu = 11, \quad p = 0.6846. \quad (88)$$

The $z = 2.33$ Ly α block contributes

$$\chi_{\text{Ly}\alpha}^2 = 2.1621, \quad \nu = 2, \quad p = 0.3392, \quad (89)$$

when evaluated with its published two-dimensional covariance.

9.2 Redshift-range stability

For the diagnostic low-redshift subset $z \leq 0.706$,

$$\chi^2 = 8.3452, \quad \nu = 5, \quad p = 0.1382. \quad (90)$$

For $0.934 \leq z \leq 1.321$,

$$\chi^2 = 2.5429, \quad \nu = 4, \quad p = 0.6370. \quad (91)$$

For $z \geq 1.484$, including the Ly α block,

$$\chi^2 = 2.7144, \quad \nu = 4, \quad p = 0.6067. \quad (92)$$

No redshift range requires a change in the acoustic projection.

9.3 Transverse and radial sectors

The six transverse measurements give

$$\chi_{D_M}^2 = 6.6976, \quad \nu = 6, \quad p = 0.3497. \quad (93)$$

The six radial measurements give

$$\chi_{D_H}^2 = 6.6394, \quad \nu = 6, \quad p = 0.3555. \quad (94)$$

The overall closure is therefore not produced by one geometrical direction compensating for failure in the other.

9.4 Residual structure

Figure 2 shows the marginal pulls

$$\delta_i = \frac{m_{\text{CPTG},i} - d_i}{\sqrt{C_{ii}}}. \quad (95)$$

These are visualization diagnostics only; correlated measurements are evaluated with their full covariance. The Ly α residuals are $+0.449\sigma$ in D_H/r and -1.263σ in D_M/r . The largest marginal residual in the complete vector remains

$$|\delta|_{\text{max}} = 1.607\sigma, \quad (96)$$

for D_H/r at $z = 0.706$.

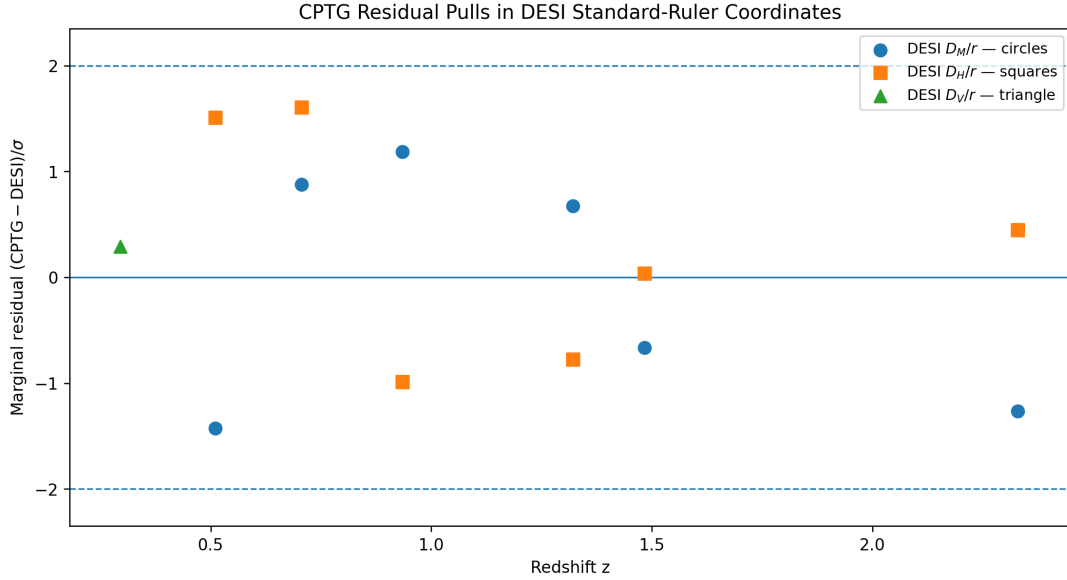


Figure 2: Marginal residual pulls for the fixed CPTG prediction relative to the DESI vector. The $z_{\text{eff}} = 2.33$ points use the joint Ly α AP+BAO measurement. The dashed lines mark $\pm 2\sigma$. Because the DESI measurements are correlated within redshift blocks, these marginal pulls are diagnostic and do not replace the covariance statistic.

9.5 Unprojected native- π negative control

The decisive internal control is to remove the acoustic projection and use the native exponent $p = \pi$, while still allowing one global normalization to optimize against the DESI vector. This gives

$$\chi_\pi^2 = 113.392792483, \quad \nu = 12, \quad (97)$$

with

$$p_\pi = 1.27 \times 10^{-18}. \quad (98)$$

The failure is therefore not removable by rescaling the native branch.

A normalization-free anisotropic shape check uses the six redshift blocks for which both transverse and radial coordinates are available. For each block define the Alcock–Paczynski ratio

$$F_{\text{AP}}(z) \equiv \frac{D_M(z)/r}{D_H(z)/r} = \frac{D_M(z)}{D_H(z)} = E(z)I(z). \quad (99)$$

For a measured pair $\mathbf{y} = (D_M/r, D_H/r)^\top$ with within-block covariance $\mathbf{C}_y$, the marginal variance of the ratio is propagated with

$$\sigma_{F,i}^2 = \mathbf{J}_i \mathbf{C}_{y,i} \mathbf{J}_i^\top, \quad \mathbf{J}_i = \left(\frac{1}{D_H/r}, -\frac{D_M/r}{(D_H/r)^2} \right)_i. \quad (100)$$

The shape-only statistic is then

$$\chi_{\text{shape}}^2 = \sum_{i=1}^6 \frac{[F_{\text{AP},i}^{\text{model}} - F_{\text{AP},i}^{\text{DESI}}]^2}{\sigma_{F,i}^2}. \quad (101)$$

The common dimensional ruler cancels identically in Eq. (99); this diagnostic therefore tests the anisotropic redshift dependence without fitting a normalization. For the fixed acoustic projection,

$$\chi_{\text{ac,shape}}^2 = 7.9034, \quad \nu = 6, \quad p = 0.2453, \quad (102)$$

while the raw native shape gives

$$\chi_{\pi,\text{shape}}^2 = 36.7557, \quad \nu = 6, \quad p = 1.96 \times 10^{-6}. \quad (103)$$

The high-redshift AP information preserves the distinction between the accepted acoustic projection and the rejected unprojected native shape.

10 One Projection Across CMB and BAO Coordinates

The calculations above establish two separate observational checks of one acoustic projection. The first is the CMB-facing endpoint and angular-scale construction. The second is the late-time DESI BAO covariance test. They should not be interpreted as two independently tuned CPTG cosmologies.

The upstream structure is

$$\mathfrak{B}_\pi \longrightarrow \mathcal{P}_{\text{ac}} \longrightarrow \mathfrak{B}_{\text{ac}}, \quad (104)$$

with the acoustic branch fixed by Eqs. (19)–(28). The observational comparison then branches only at the level of reported coordinates:

$$\mathfrak{B}_{\text{ac}} \longrightarrow \begin{cases} \text{CMB angular acoustic coordinate,} \\ \text{BAO distance/ruler coordinates.} \end{cases} \quad (105)$$

The CMB interface supplies the observed angular scaffold used in Eq. (54). DESI then tests the downstream vector generated from that already-fixed quantity. No DESI measurement is used to redefine the acoustic exponent, the native coefficient, the present acoustic rate normalization, or the CMB-derived dimensionless BAO normalization.

This is the same projection logic that underlies the CPTG Hubble-bridge construction: an acoustic comparison rate and an observational distance coordinate are different readouts of one projected geometry rather than independent expansion laws [4, 5]. The standard-ruler calculation strengthens that architecture because the projected response is tested not only at a present endpoint but across multiple DESI redshifts and in both radial and transverse distance channels.

The coordinate identity introduced in Section 5 should be understood in this context. It does not enlarge the theory with another transport law. Once $H_{\text{ac}}(z)$ has been fixed by the acoustic projection, D_H , D_M , D_V , and the corresponding ruler ratios follow from their definitions. The difference between a local radial response and an integrated transverse response is therefore a consequence of how the same projected branch is observed.

The strongest result of the present paper is accordingly not the isolated numerical value $r_{\text{ac}} = 148.6219 \text{ Mpc}$. That dimensional number is a useful representation of the comparison coordinate, but the direct observational closure is dimensionless:

$$A_{\text{CPTG}} = 29.8935221443 \pm 0.0089020704, \quad (106)$$

followed by the zero-fit DESI vector in Eqs. (61)–(63). The same fixed acoustic projection therefore survives translation between the CMB and BAO observational interfaces without a DESI-specific recalibration.

11 Observational Comparison Summary

Table 1 collects the principal scalar comparisons. Standardized residuals are reported only where a one-dimensional comparison with a defined uncertainty is appropriate.

Comparison	CPTG	Comparison coordinate	Residual	Standard σ
Acoustic H_0 [$\text{km s}^{-1} \text{ Mpc}^{-1}$]	67.4777967351	Planck ΛCDM : 67.4 ± 0.5	+0.0777967351	+0.156 σ
CMB-to-DESI normalization A	$29.8935221443 \pm 0.0089020704$	$29.8295509892 \pm 0.0816008671$	$+0.0639711551 \pm 0.0820850069$	+0.779 σ
Equivalent acoustic coordinate [Mpc]	$148.621869963 \pm 0.044258497$	$148.940597948 \pm 0.407437643$	$-0.318727985 \pm 0.409834414$	−0.778 σ
$\text{Ly}\alpha$ D_H/r , $z = 2.33$	8.62962731	8.600 ± 0.066	+0.02962731	+0.449 σ
$\text{Ly}\alpha$ D_M/r , $z = 2.33$	38.90310459	39.32 ± 0.33	−0.41689541	−1.263 σ

Table 1: Principal scalar comparisons. The Planck H_0 row is an external endpoint comparison. The CMB-to-DESI normalization uses the fixed CPTG acoustic shape. The final two rows show the $\text{Ly}\alpha$ AP+BAO high-redshift anchor; their joint significance is evaluated with the published correlation rather than from the marginal pulls alone.

The covariance-level tests are summarized in Table 2. These multivariate results are

reported with χ^2 , degrees of freedom, and p -value rather than converted into a one-dimensional Gaussian sigma.

Test	χ^2	ν	p	Diagnostic
DESI zero-fit CPTG	13.6025	13	0.4024	max marginal pull 1.61σ
Ly α block	2.1621	2	0.3392	published 2D covariance
Transverse D_M/r sector	6.6976	6	0.3497	no retuning
Radial D_H/r sector	6.6394	6	0.3555	no retuning
Low redshift, $z \leq 0.706$	8.3452	5	0.1382	diagnostic split
Intermediate redshift	2.5429	4	0.6370	$0.934 \leq z \leq 1.321$
High redshift, $z \geq 1.484$	2.7144	4	0.6067	includes Ly α
Raw native- π control, fitted scale	113.3928	12	1.27×10^{-18}	one fitted normalization

Table 2: Covariance-level closure and robustness summary. The CPTG rows use the fixed acoustic projection without DESI retuning. The final row is the unprojected native negative control with one optimized normalization.

The high-redshift information is fully consistent with the fixed CPTG projection and preserves the discriminatory power of the negative control.

12 Falsifiability, Limitations, and Scope

The result in this paper is intentionally narrower than a complete cosmological theory test. Its strength comes from keeping the projected geometry fixed and stating the comparison boundary explicitly.

The zero-fit standard-ruler closure would fail if future BAO or AP+BAO measurements, evaluated in the same dimensionless coordinates with their published covariance, become incompatible with Eqs. (61)–(63). No retuning of p_{ac} , $\mathcal{G}_T$, or A_{CPTG} is permitted in that test. It would likewise fail if radial and transverse measurements require mutually incompatible projected histories.

The CMB boundary used here contains one directly constrained quantity, $\theta_\star^{\text{obs}}$, and one Planck-derived endpoint, $z_\star^{\text{Pl}}$. They are comparison inputs rather than native CPTG invariants. Their use does not add an intermediate evolution model to CPTG: once the relevant boundary coordinates and the fixed acoustic projection are specified, the comparison coordinate follows.

The dimensional r_{ac} is likewise a comparison-coordinate representation. The DESI denominator r_d is retained as the conventional label of the published survey coordinate; this paper does not claim that the CPTG r_{ac} is microscopically identical to the standard-model drag horizon. The observational test is performed on the dimensionless DESI ratios.

The high-redshift constraint is the joint Ly α AP+BAO result at $z_{\text{eff}} = 2.33$, while the lower-redshift block uses the public DESI DR2 BAO covariance. The controlling covariance is assembled from these two published DESI likelihood components.

Finally, the redshift and sector partitions in Section 9 are diagnostics of stability, not independent likelihood datasets. Marginal pull plots likewise do not replace the covariance statistic.

Within these boundaries, the demonstrated result is specific and falsifiable: one fixed CPTG acoustic projection, combined with the declared CMB comparison boundary, generates a zero-DESI-fit vector accepted by the DESI BAO and Ly α AP+BAO constraints.

13 Discussion

The standard-ruler test developed here separates three layers that are often blended in conventional parameter inference: the upstream theoretical geometry, the acoustic comparison projection, and the survey coordinate in which the result is reported.

CPTG fixes the native branch first. The exact acoustic projector then changes the response exponent while preserving the inherited native transport coefficient. The CMB comparison boundary supplies the angular normalization, after which the result is carried into the DESI D_M/r , D_H/r , and D_V/r coordinates. No DESI-specific dimensional ruler is fitted.

The Ly α AP+BAO result provides the high-redshift test. At $z_{\text{eff}} = 2.33$, DESI measures D_H/r_d to 0.77% and D_M/r_d to 0.84%, with nonzero covariance [7]. The fixed CPTG predictions at that redshift differ from the measured central values by only $+0.449\sigma$ and -1.263σ , respectively. The two-dimensional high-redshift block gives $p = 0.3392$.

Across the full vector, the zero-fit result is

$$\chi^2 = 13.6025, \quad \nu = 13, \quad p = 0.4024.$$

The transverse and radial sectors remain independently acceptable, with $p = 0.3497$ and $p = 0.3555$. The CMB-derived normalization differs from the DESI-preferred normalization by 0.779σ .

The negative control is equally important. When the acoustic projection is removed and the native branch is allowed to optimize one global normalization, the DESI vector gives $p = 1.27 \times 10^{-18}$. A normalization-free anisotropic-shape test reaches the same qualitative conclusion: the acoustic projection is accepted, whereas the unprojected native shape is strongly disfavored.

The result therefore does not depend on treating the BAO denominator as an independently calibrated CPTG rod. The stronger statement supported here is narrower: one fixed CPTG native-to-acoustic projection can be translated from the declared CMB boundary into the DESI standard-ruler coordinates, including the high-redshift AP information, without a DESI-specific scale fit.

14 Conclusion

This paper tested the cosmic standard ruler in Curvature Polarization Transport Gravity from the CMB acoustic comparison boundary to the DESI DR2 distance coordinates.

The calculation begins from one native geometric π -branch and its exact acoustic projection. The present acoustic comparison rate,

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

is derived upstream and differs from the Planck base- Λ CDM CMB-inferred comparison value $67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ by 0.156σ .

Using the measured CMB acoustic angle and the adopted Planck-derived decoupling endpoint gives

$$A_{\text{CPTG}} = 29.8935221443 \pm 0.0089020704.$$

For the DESI vector, the conditionally preferred normalization is

$$A_{\text{DESI}} = 29.8295509892 \pm 0.0816008671,$$

a difference of 0.779σ .

The controlling comparison contains no DESI-fitted parameter. The 13-observable covariance gives

$$\chi^2 = 13.6024621, \quad \nu = 13, \quad p = 0.4024.$$

The largest marginal residual remains 1.61σ . The result survives redshift-block removal, broad redshift partitions, and separate radial and transverse tests.

The unprojected native branch fails the same data even when one global normalization is optimized:

$$\chi_{\pi}^2 = 113.3928, \quad \nu = 12, \quad p = 1.27 \times 10^{-18}.$$

The success of the acoustic branch is therefore not an arbitrary rescaling of the native curve.

The standard-ruler result does not add a new CPTG law and does not identify the CPTG acoustic comparison coordinate microscopically with the standard-model drag horizon. It demonstrates that the established native-to-acoustic projection can be carried from the CMB boundary into the radial, transverse, and isotropic DESI coordinates, including the $\text{Ly}\alpha$ AP+BAO high-redshift anchor, without a DESI-specific dimensional ruler fit.

Data, Code, and Evidence Availability

The lower-redshift comparison uses the public DESI DR2 Gaussian BAO mean vector and covariance through $z = 1.484$ [2]. The high-redshift block uses the joint Ly α AP+BAO constraint at $z = 2.33$,

$$D_H/r_d = 8.600 \pm 0.066, \quad D_M/r_d = 39.32 \pm 0.33, \quad \rho = 0.225,$$

published by the DESI Collaboration [7]. The analysis covariance combines the public lower-redshift block with the published 2×2 high-redshift covariance.

The reproducibility record consists of the fixed CPTG prediction vector, the 13-observable covariance calculation, the conditional normalization diagnostic, leave-one-redshift-block-out tests, redshift and sector partitions, the anisotropic shape check, the unprojected native- π negative control, and the two publication figures. The complete standalone evidence chain is archived as:

`CPTG_Cosmic_Standard_Ruler_CMB_to_DESI_Evidence_Package.zip`

The archive contains the publication TeX source, both reproducibly generated figures, frozen Planck and DESI input snapshots, the executable numerical reproduction script `CPTG_DESI_BAO_BlockCovariance.py`, the figure-generation script `CPTG_DESI_BAO_MakeFigures.py`, summary and per-observable output tables, Windows run commands, a clean reproduction log, source-provenance notes, and a SHA-256 manifest. The controlling numerical reproduction is offline and fail-closed: it verifies the archived input hashes, derives I_* , A_{CPTG} , and $\sigma_{A,\text{CMB}}$ from the frozen CMB comparison inputs, and then assembles the frozen 11-observable lower-redshift DESI block with the joint Ly α AP+BAO 2×2 block used here. No network retrieval is used by the controlling calculation. Supporting CPTG research materials and public benchmark implementations are maintained in the CPTG repository [6].

References

- [1] N. Aghanim et al. [Planck Collaboration], *Planck 2018 results. VI. Cosmological parameters*, Astronomy & Astrophysics **641**, A6 (2020), DOI: 10.1051/0004-6361/201833910, arXiv:1807.06209.
- [2] M. Abdul Karim et al. [DESI Collaboration], *DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints*, Physical Review D **112**, 083515 (2025), DOI: 10.1103/tr6y-kpc6, arXiv:2503.14738.
- [3] Carter L Glass Jr, *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, (2026), DOI: 10.5281/zenodo.22441039.
- [4] Carter L Glass Jr, *CPTG Geometric Pi Branch: Cosmological Comparison Coordinates*, (2026), DOI: 10.5281/zenodo.22426857.
- [5] Carter L Glass Jr, *CPTG and the Hubble-Tension Bridge*, (2026), DOI: 10.5281/zenodo.22429502.
- [6] CPTG, *Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity*, <https://github.com/CLG2025/CPTG>.
- [7] DESI Collaboration et al., *DESI DR2 Results IV: Alcock–Paczynski Measurements from the Lyman Alpha Forest and Cosmological Constraints*, (2026), arXiv:2607.27410.
- [8] B. Popovic and M. Sullivan, *On the Stability of H_0 and the Inverse Distance Ladder*, Monthly Notices of the Royal Astronomical Society **550**, stag1267 (2026), DOI: 10.1093/mnras/stag1267.
- [9] J. A. Nájera, I. Banik, H. Desmond, and V. Kalaitzidis, *Comparing Measures of the Hubble and BAO Tensions in Λ CDM and Possible Solutions in $f(Q)$ Gravity*, Galaxies **14**, 19 (2026), DOI: 10.3390/galaxies14020019.
- [10] A. Pérez-Fernández et al. [DESI Collaboration], *Fiducial-Cosmology-dependent systematics for the DESI 2024 BAO Analysis*, Journal of Cosmology and Astroparticle Physics **2025**(01), 144 (2025), DOI: 10.1088/1475-7516/2025/01/144, arXiv:2406.06085.